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Hubble Constant Assessment Report

The aim of this assessment was to determine the Hubble constant which is defined by the equation:

$$Hubble\ Constant = Recession\ Velocity / Distance\ from\ Galaxy \quad (1)$$

The recession velocities have been given in the data however to obtain the distances from galaxies we use the average distance of cephids in a galaxy:

$$M = m - 5\log(d) + 5 - A \quad (2)$$

Where M = absolute magnitude of the cephids, m = apparent magnitude of the cephids, A = the extinction of the cephids and d is the distance in parsecs from the cephids.

We are given a data file containing data for m and A however to get the absolute magnitude of the cephids we use:

$$M = \alpha \log(P) + \beta \quad (3)$$

Where α and β are constants to be found and P is the pulse period of a cephid.

To find alpha and beta a plot of absolute magnitudes of cephids in the Milky Way against log(Pulse period) will give a gradient of alpha and a y intercept of beta.

In the data set given values for parallax, pulsation period, apparent magnitude and extinction. After calculating the distances to each cephid using the equation:

$$Distance(pc) = 1000/Parallax \quad (4)$$

Values for M can be calculated using equation (2) and then be used with the logarithm of the Pulsation period to obtain estimates for alpha and beta.

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Step 1 – Obtain values for *α and β*

The aim of step one was to calculate values for alpha and beta in the equation M = α log(P) + β given data on cephids in the Milky Way.

Firstly, to calculate alpha and beta a set of parallaxes of cephids along with the uncertainties was used in the equation:

$$distance = 1000/parallax$$

The distance uncertainty was propagated analytically using the derivative of the distance equation multiplied by the uncertainty:

$$distance\ uncertainty = (1000/parallax^2) * parallax\ uncertainty$$

Using these sets of distances a set of absolute magnitudes can be calculated with equation *(2)*.

Along with this set of absolute magnitudes an analytical and numerical set of uncertainties was calculated which both produced very similar results however I decided to use the numerical results in future calculations since it can account for random variation in the parameters such as extinction or parallax.

Using the calculated M values a plot was made against log(P) which was given by the data set.

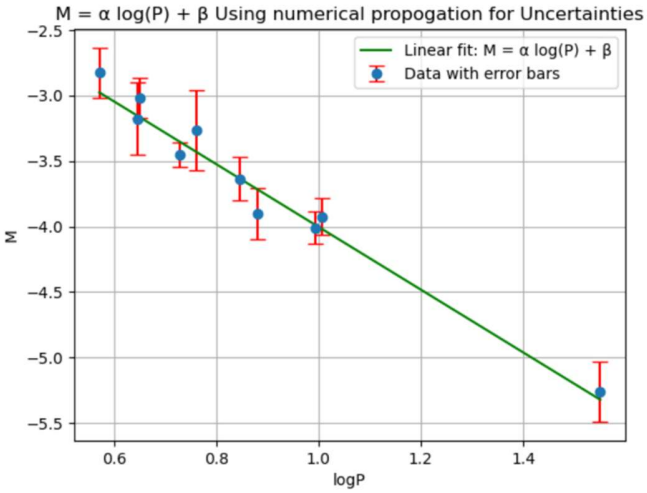


Fig 1 shows a plot of absolute magnitudes (M) against log(P) with a linear fit (green line) which uses the parameters calculated to fit the data. The blue circles are the measured values for the M and log(P) with the uncertainty in absolute magnitude shown in red.

The linear fit was found by SciPy's *curve_fit* function which finds the optimal parameters to fit the observations and gave values for the slope and y intercept which are equal to alpha and beta respectively due to the linear nature of the equation $M = \alpha \log(P) + \beta$. Using curve fit will also output the covariance of parameters which is used to calculate the uncertainties in the final values for alpha and beta.

Alpha = -2.39 ± 0.23 Beta = -1.61 ± 0.20

To verify the fit of the data a Chi^2 test was done, and the fit had a value of *Chi^2 = 4.41* and a reduced *Chi^2 = 0.55068*. Relating this to the expected *Chi^ 2 = 8 ± 4.0* with 8 degrees of freedom, we see that my data does fit within the range showing that the relation is a good fit to the data which can also be seen in the graph as the fit line (green line) passes through most of the data points error bars. This proves that the values for alpha and beta fit the data accurately so we can accept them as values confidently.

One way to possibly improve these values is to perform a correlation correction before calculating the parameters, this may be useful as the value of beta will change if alpha changes however after correlation correction, they will not be dependent on each other. This may influence the results but likely to small to make any significant difference so was not done in the code.

Step 2 – Find distance to ngc4527

The aim of step 2 is to use the calculated values of alpha and beta to find values for absolute magnitudes of cephids outside of the milky way and then to use these magnitudes to find the distance to the cephids and estimate the distance to the galaxy they are in.

Now a value for alpha and beta has been found, data containing log(P) and apparent magnitudes of cephids in the galaxy ngc4527 can be used to find M for each cephid. These calculations were done numerically to obtain uncertainties for these values. The M values are used in a rearranged equation (2):

$$d = 10^{((m+5-M-A) / 5)}$$

which gives distance values of each cephid along with the numerical uncertainties for each. Any time a numerical uncertainty is calculated gaussian distributions of uncertainties must be assumed.

After calculating the distances to each cephid I checked the data for any outliers in the data.

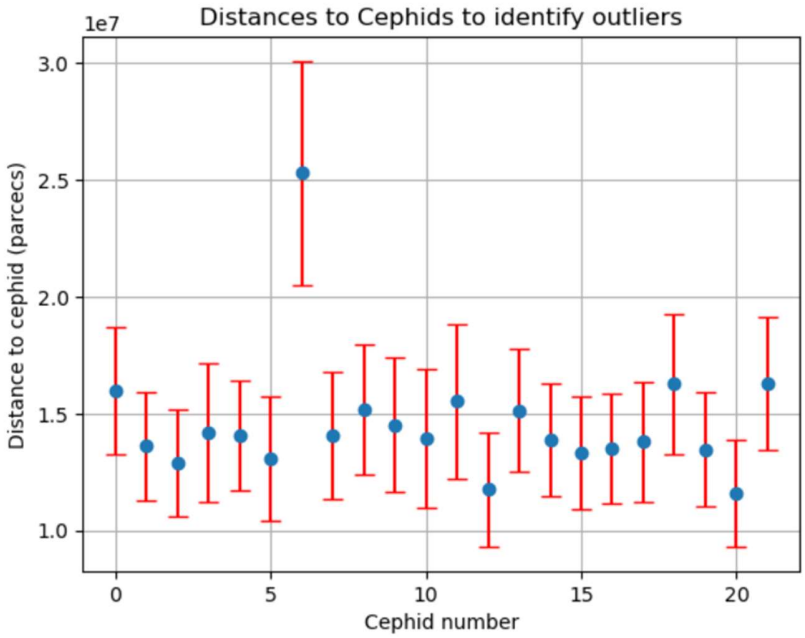


Fig 2 shows the distances of each cephid from earth in blue with their uncertainties in red. The data shows one clear outlier (Cephid 6) which will be proven to be an outlier.

To verify if the presence of an outlier the Inter Quartile Range method was used (outliers are outside the range of 1.5*IQR from the mean). After performing the test cephid 6 was determined to be an outlier so was removed from the data set for future calculations.

With the distances of each cephid calculated the distance of the galaxy can be estimated by the average distance of the cephids with that galaxy. To get an accurate estimate, the NumPy function *average* was used which can perform weighted means using the uncertainties in each distance for a more accurate estimate. The uncertainties in this distance were calculated by the standard error of the cephid distance.

Distance of ngc4527 without removing outliers: 15.1 ± 0.6 Mpc

Distance of ngc4527 after removing outliers: 14.2 ± 0.3 Mpc

From these results we can see after removing the outlier the uncertainty is smaller which is why this value will be used in further calculations.

Using this distance an estimate for the Hubble constant can be found using the known recession velocity of the galaxy (1152 km/s) and equation (1) with the uncertainty propagated analytically from the uncertainty in distance.

H0 = 81.1 ± 1.5 (km/s)/Mpc

Step 3

The aim of the final step is to find a value of the Hubble constant using recession velocity and distance data of 7 galaxies paired with the data from step 2.

In order to find the Hubble constant a plot of recession velocity against galaxy distance from the 8 galaxies should show a linear relation where the slope is the Hubble constant:

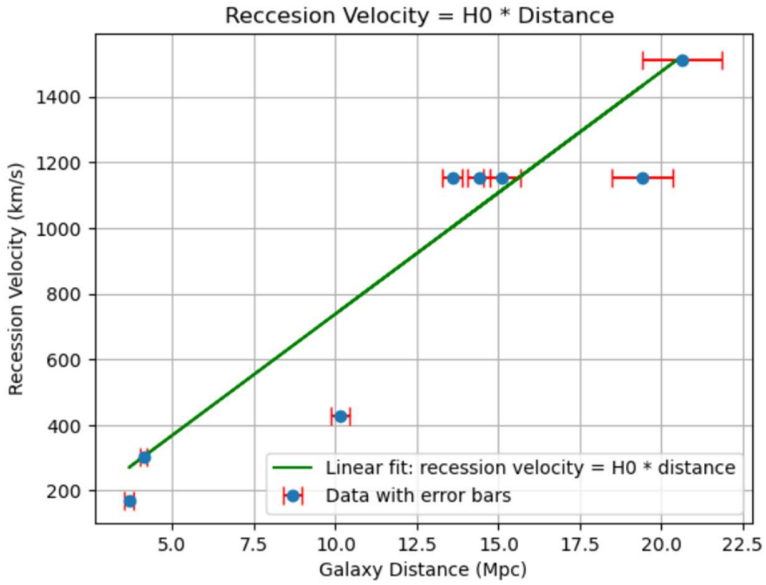


Fig 3 shows a plot of recession velocity against galaxy distance where the green line is a linear fit of the data, the gradient of this line should be an estimate of the Hubble constant. The blue circles represent the data for recession velocity of the galaxies with the red representing the uncertainty in the distance to each galaxy.

H0 = 73.9 ± 6.1 (km/s)/Mpc

This value was found using the curve_fit function which found optimal parameters for the model along with the covariance which can be used to find the uncertainty of the parameters. However, after finding Chi^2 for this model we get a value of *Chi^2 = 413.8* which does not fit within the range of expected *Chi^2 = 7 ± 3.7*

This result of a very large Chi^2 suggests that the data used does not fit the model, one possible reason for this could be due to underestimated uncertainties for the galaxy distance. This underestimated uncertainty can be seen in the graph of the data as the linear fit (green line) crosses through very few of the data points error bars. If the error bars were larger (not underestimated) than the line may cross through more of the lines meaning the fit would match the data more accurately providing a smaller Chi^2.

Due to the very high Chi^2 value I calculated a different H0 value by calculating an estimate of H0 for each of the galaxies individually paired with the uncertainties calculated numerically assuming a Gaussian distribution of uncertainties. Then using the multiple H0 values calculated a weighted mean H0 with an uncertainty of the standard error of the data providing a value of H0 = 69.1 +- 5.2 (km/s)/Mpc. This value is similar to the value calculated from the graph method which supports the final value for the Hubble constant.

Final value of Hubble Constant = 73.9 ± 6.1 (km/s)/Mpc